Journal of Food Protection 88 (2025) 100665

Contents lists available at ScienceDirect

Journal of Food Protection

journal homepage: www.elsevier.com/locate/jfp

Research Paper

Investigating the Dynamics of *Salmonella* Contamination During Commercial Pork Harvest Using Qualitative and Quantitative Detection Methods

J.A. Brown, S.C. Ricke *

Meat Science and Animal Biologicals Discovery (MSABD) Program, Department of Animal and Dairy Sciences, University of Wisconsin-Madison, Madison, WI 53706, USA

ARTICLE INFO

Keywords:
Harvest
Pork
Prevalence
Quantitative
Salmonella

ABSTRACT

Animals can be reservoirs for pathogens such as *Salmonella*, creating the potential for cross−contamination during harvest. Antimicrobials are used to limit this risk, but validation of interventions in commercial settings can be challenging. The objective of this study was to develop a bio-map to evaluate the presence and concentration of *Salmonella* during harvest in commercial market hog facilities. Carcass swabs were collected from three commercial market hog facilities, five times a day for five days, at nine locations throughout harvest from exsanguination to chilling ($N = 675$). At each sampling interval, a single carcass was tracked through harvest by swabbing the ham, shoulder, and jowl. Samples were analyzed for quantitative and qualitative *Salmonella* using Gene-Up Quant ($\log_{10}$ CFU/swab; limit of quantification (LOQ) = 1 $\log_{10}$ CFU/swab) and Gene-Up SLM2 (present/absent; limit of detection (LOD) = 1 CFU/swab), respectively. Prevalence data were analyzed in R Studio using linear mixed effect models, with carcass as a repeated measure, and ANOVA with means separated by Tukey's HSD ($p \leq 0.05$). Due to the limited number of quantitative results above the LOQ ($n = 7$; 1.15–1.56 $\log_{10}$ CFU/swab), statistical analyses were not performed. *Salmonella* prevalence was impacted by facility ($p = 0.006$), location ($p < 0.001$), and the interaction between facility and location ($p = 0.004$). Across all facilities, *Salmonella* prevalence was highest on incoming pigs and decreased most significantly after external decontamination. These results indicate that while concentrations may be low, *Salmonella* contamination is prevalent throughout the pork harvest process, highlighting the need for effective detection systems to develop and evaluate process control strategies.

Salmonella is one of the leading causes of bacterial foodborne illness in the United States and worldwide (Scallan et al., 2011). It is estimated that nontyphoidal *Salmonella* causes 1.35 million illnesses in the United States each year (Centers for Disease Control and Prevention, 2019). Due to the vast number of *Salmonella* serotypes with varying levels of pathogenicity, the infectious dose is reported to range from 1,000 to 100,000 cells (Kothary & Babu, 2001; Li et al., 2012). The most recent attribution data from 2022 reported that more than 75% of all salmonellosis cases were attributed to seven food categories, including several animal-sourced commodities such as chicken (19.7%), pork (11.9%), beef (6.9%), and turkey (5.2%) (Interagency Food Safety Analytics Collaboration, 2024). Despite considerable efforts to reduce the incidence of salmonellosis, incidence rates have remained stagnant over the last two decades, requiring regulatory officials to reassess their approach to achieving the Healthy People 2020, now 2030, target of a 25% reduction (U.S. Department of Agriculture Food Safety and Inspection Service, 2013; U.S. Department of Health and Human Services, 2022).

Swine and other livestock animals frequently serve as reservoirs for potentially pathogenic microorganisms, including *Salmonella* (World Health Organization, 2018). The presence of these microorganisms on live animals creates a potential food safety risk due to the likelihood of cross-contamination during harvest (Wheatley et al., 2014). To mitigate this risk, the pork industry uses effective sanitary dressing procedures to prevent contamination and various types of intervention strategies to reduce the concentration and/or prevalence of target microorganisms (Spescha et al., 2006; U.S. Department of Agriculture Food Safety and Inspection Service, 2019). Common intervention strategies employed during pork harvest include the use of mechanical action to remove external contaminants (i.e. dehairing and polishing), the application of heat to thermally inactivate microbial contaminants (i.e. scalding and singeing), and the application of antimicrobials to chemically inactivate microbial contaminants (i.e. wash cabinets) (Gill et al., 1995; Gonzalez et al., 2023). These interventions can be used independently but are more frequently utilized in combination as a multihurdle approach (Gragg & Brashears, 2014).

* Corresponding author at: Department of Animal and Dairy Sciences, Office 2124 MSABD, University of Wisconsin - Madison, 1933 Observatory Drive, Madison, WI 53706, USA.
E-mail address: sricke@wisc.edu (S.C. Ricke).

https://doi.org/10.1016/j.jfp.2025.100665
Received 18 June 2025; Accepted 2 November 2025
Available online 6 November 2025
0362-028X/© 2025 The Author(s). Published by Elsevier Inc. on behalf of International Association for Food Protection.
This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

The implementation of pathogen performance standards as part of the "Pathogen Reduction; Hazard Analysis and Critical Control Points Systems" (PR/HACCP Rule; 61 FR 38806) was the first major regulatory policy specifically designed to reduce the occurrence of *Salmonella* in meat and poultry products (U.S. Department of Agriculture Food Safety Inspection Service, 1996). This rule established performance standards for beef, pork, and poultry carcasses at slaughter, as well as ground beef, turkey, and chicken (9 CFR §310.25). Despite the high rate of attribution associated with pork products, performance standards for fresh pork sausage were never finalized and pork carcass sampling was halted in 2012 after finding the prevalence to be very low (U.S. Department of Agriculture Food Safety and Inspection Service, 2011). The poultry industry was the first to see new *Salmonella* performance standards implemented in 2016 (81 FR 7285) and later revised in 2018 (83 FR 56046) (U.S. Department of Agriculture Food Safety Inspection Service, 2016, 2018). These policies created the framework for future implementation of *Salmonella* performance standards in beef and pork processing establishments proposed in 2019 (84 FR 57688) and 2022 (87 FR 8774), respectively (U.S. Department of Agriculture Food Safety Inspection Service, 2019a, 2022). The proposed performance standards for the pork industry set the maximum number of allowable positive samples, within the 52−week rolling window, at 13 (25%) and 6 (11.5%) for raw comminuted pork and raw intact or nonintact pork cuts, respectively (87 FR 8774). While no final rules have been implemented for the pork industry, the potential implications of regulatory standards highlight the importance of understanding and controlling *Salmonella* contamination within pork processing facilities.

The food industry has historically relied on culture-based methodologies as the "gold standard" to test for microbial contamination during processing and in finished products (Neyaz et al., 2024; Priyanka et al., 2016). Although culture-based methodologies have been a reliable form of pathogen detection for several decades, their limitations related to labor requirements and extended time to result (~5 days) led to the emergence of molecular-based detection systems (Bell et al., 2016; U.S. Department of Agriculture Food Safety and Inspection Service, 2021; Yang et al., 2025). Molecular microbial detection technologies offer a rapid and sensitive alternative to the traditional culture-based methods, using DNA to identify target organisms rather than relying on the microorganism's ability to grow in a controlled environment (Law et al., 2015).

Qualitative and quantitative molecular-based detection systems are both currently available for use in the food industry. Qualitative assays have largely replaced the previous "gold standard" due to their reliability, sensitivity, and rapid time to results. Through the use of an overnight culture-based enrichment, these assays are highly sensitive (1 CFU/g) and applicable in a variety of sample matrices (Park et al., 2014). In contrast, quantitative detection assays have been developed both with and without enrichment to quantify the levels of target pathogens between 1 and 7 $\log_{10}$ CFU/sample. Assays that perform a nonenrichment quantitative assessment of target pathogens are generally more accurate but also more susceptible to environmental interference. Quantitative assays that include an enrichment step are slightly more robust; however, these assays can be limited in that they can only estimate the actual concentration using a defined standard curve (Schmidt et al., 2024). While quantitative detection assays tend to provide more detailed and actionable results than qualitative assays, limitations associated with sensitivity and reliability currently limit their application in food processing environments (Bell et al., 2016; Bueno López et al., 2022; Sidstedt et al., 2020).

The objective of this research was to investigate *Salmonella* contamination within three commercial pork harvest facilities using qualitative and quantitative *Salmonella* detection assays to generate microbiological maps, or bio-maps, of each process. Each carcass swab was analyzed using qualitative and quantitative molecular detection systems, and results were compared to assess the performance of each technology under real-world processing conditions. While process flows were similar across all harvest facilities, there were certain aspects related to facility design and equipment usage that made each facility unique. The generation of individual *Salmonella* bio-maps for each facility provided the opportunity to evaluate individual processing steps and pieces of equipment to determine their impact on *Salmonella* contamination. These results will provide the pork industry with a valuable baseline measurement for *Salmonella* contamination at various stages of harvest. Furthermore, understanding the impact of individual processing steps and equipment will aid in the development of targeted intervention strategies and facility/equipment design. Finally, evaluating the commercial applicability and utility of quantitative molecular detection systems will provide insight into method optimization and effective application strategies within the food industry.

Materials and methods

Sample collection. Sample collection occurred at three commercial market hog processing facilities (A, B, and C) in the fall of 2022. Nine sampling locations were identified throughout harvest: (A) exsanguination, (B) gambrel table, (C) after shavers, (D) after opening, (E) after bunging, (F) after evisceration, (G) after split, (H) prewash cabinet, and (I) postwash cabinet (Fig. 1). The antimicrobial wash cabinet located between the pre- and postwash cabinet sampling locations applied peracetic acid (PAA) at all facilities. Each facility was sampled five times a day for five consecutive days to account for day-to-day and shift variation. During each sampling interval, a carcass was identified at exsanguination and subsequently followed through harvest and sampled at the nine specified locations to longitudinally assess the impact of individual processing steps. Samples were collected in accordance with the United States Department of Agriculture Food Safety Inspection Service (USDA FSIS) guidelines for swabbing pork carcasses, swabbing the ham, shoulder, and jowl at each locations with a sponge stick swab presoaked in 10 mL of buffered peptone water (BPW; 3 M, St. Paul, MN, USA) (U.S. Department of Agriculture Food Safety and Inspection Service, 2005). This sampling scheme resulted in 25 carcasses being sampled per facility (5 carcasses/day) for a total of 225 samples at each facility over the course of one week. In total, 75 carcasses were sampled, and 675 swabs were collected across all three facilities. After collection, samples were stored at 4 °C and shipped overnight on ice packs to an ISO 17025 accredited laboratory, where each sponge swab was analyzed qualitatively and quantitatively for *Salmonella* as described in the following sections.

Quantitative *Salmonella* analysis. Upon arrival at the laboratory, carcass swabs were weighed, combined with 5 × volume to mass of BPW, and homogenized for 1 min in a paddle blender. A 40 mL portion of each homogenate was transferred to a sterile 50−mL conical tube and analyzed for quantitative *Salmonella* using the Gene-Up® Quant *Salmonella* Test Kit (bioMerieux, Salt Lake City, UT, USA) according to manufacturer instructions (AOAC 061801). The remaining homogenates were retained for qualitative *Salmonella* analyses as described below. Briefly, for the quantitative analysis, the aliquoted homogenates were centrifuged for 10 min at 500 × g to pellet large debris. The resulting supernatants (25 mL) were transferred to new conical tubes and centrifuged for a second time (10 min at 4300 × g) to pellet bacterial cells. The resulting supernatants were decanted, and the pellets were resuspended in 600 μL of lysis buffer (Nuclei Lysis Solution; Promega, Madison, WI, USA) and vortexed to fully resuspend. The entirety of the resulting solutions (600 μL) was subsequently transferred to a 1.5−mL microcentrifuge tubes and incubated in a heating block for 5 min at 80 °C. After incubation, samples were cooled on ice for 2 min before 200 μL of Protein Precipitation Solution (Promega, Madison, WI, USA) was added, and samples were vortexed to homogenize. Homogenized samples were incubated on ice for 5 min and centrifuged at 16,000 × g for 3 min. A 500 μL aliquot of each

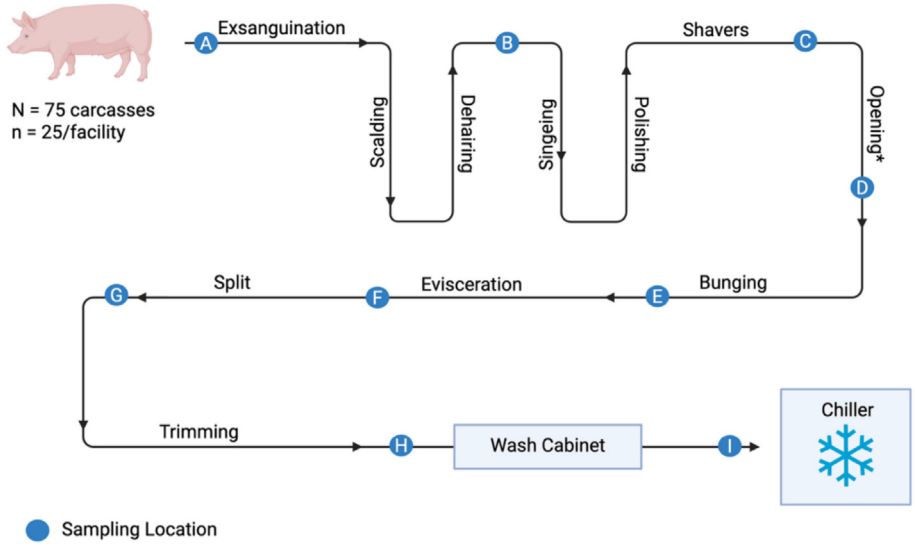


Figure 1. Diagram of a pork harvest facility with sampling locations indicated. For the purposes of this flow diagram, the harvest process at each facility began after stunning. Sampling locations are indicated by blue circles, and process flow is indicated by the directionality of the arrows. For each set of swabs, one carcass was identified at exsanguination, subsequently tracked through the remainder of processing, and sampled at each of the following locations: (A) exsanguination, (B) gambrel table, (C) after shavers, (D) after opening*, (E) after bunging, (F) after evisceration, (G) after splitting, (H) prewash cabinet, and (I) postwash cabinet. Each facility was sampled five times a day for five consecutive days, resulted in 25 carcasses sampled (5 carcasses/day) and a total of 225 swabs collected per facility (N = 675). *Opening and bunging were performed in reverse order, bunging then opening, in facility A. Created in BioRender. Brown, J. (2025) https://BioRender.com/ta3qzx1.

supernatant was transferred to a tube containing 600 μL of 95 to 100% ethanol, inverted to precipitate DNA, and centrifuged at 16,000 × g for 3 min. After centrifuging, the supernatants were decanted and tapped on absorbent paper to remove excess ethanol. The pellets were resuspended in 50 μL of prewarmed (80 °C) DNA Rehydration Solution (Promega, Madison, WI, USA) and vortexed to fully mix. Finally, 5 μL aliquots of resuspended DNA were transferred to thawed Gene-Up® Quant *Salmonella* PCR tubes (bioMerieux, Salt Lake City, UT, USA), vortexed, briefly centrifuged, and loaded into the Gene-Up® instrument (bioMerieux, Salt Lake City, UT, USA) for amplification and detection.

Estimated units per swab were calculated according to the manufacturer's linear regression model described in Schmidt et al. (2024) and reported as CFU/swab. The limit of quantification (LOQ) for the Gene-Up® Quant *Salmonella* assay was 1 $\log_{10}$ CFU/swab; therefore, only values >1 $\log_{10}$ CFU/swab were reported. Samples not yielding valid quantitative results, due to either assay inhibition or laboratory error, were labeled as "invalid" and removed from quantitative analyses. Because the quantitative and qualitative *Salmonella* analyses were performed independently for each sample, invalid quantitative results did not impact qualitative analyses.

Qualitative *Salmonella* analysis. The remaining portion of homogenates (~100 mL) from the previous step were qualitatively analyzed using the Gene-Up® *Salmonella* 2 kit (bioMerieux, Salt Lake City, UT, USA) following the manufacturer's instructions. Briefly, the swab sponges were homogenized in BPW and incubated for 18 to 24 h at 42 ± 1 °C. After incubation, sample bags were mixed by hand and 20 μL aliquots were transferred to Gene-Up® SLM2 lysis tubes. Lysis tubes were then vortexed at 2200 rpm for 5 min. After lysis was complete, 10 μL of each lysate was transferred to a Gene-Up® SLM2 PCR tube. Finally, PCR tubes were briefly centrifuged (10 s) and loaded into Gene-Up® Instrument (bioMerieux, Salt Lake City, UT, USA). Results were interpreted by the instrument and reported as either present or absent, with a limit of detection (LOD) of 1 CFU/swab.

Statistical analyses. All microbial analyses were performed in the 4.4.1 Version of R Studio (R Core Team, 2024). Quantitative *Salmonella* data (CFU/swab) were logarithmically transformed ($\log_{10}$ CFU/swab); however, due to the large majority of samples reported below the LOQ or invalid, no statistical analyses were performed. Rather, the combined quantitative and qualitative results have been summarized in Table 1. Qualitative *Salmonella* data were transformed into binary values based on the presence (1) or absence (0) of *Salmonella* and reported as prevalence, or percentage of positive samples. Qualitative results were then analyzed altogether as a linear mixed effect model with facility (k = 3), sampling location (k = 9), and their interaction as fixed effects and carcass as a random effect to account for the repeated measure. Fixed and random effects were determined using a Type III Analysis of Variance with Satterthwaite's method. Estimated marginal means and standard errors were calculated using the emmeans package in R Studio with means separated by Tukey's HSD (Searle et al., 1980). Results were considered significant at or below a p value of 0.05.

Results

A total of 675 swab samples were collected and analyzed for the presence/absence and quantitative level of *Salmonella*. Excluding those samples removed as a result of laboratory error or invalid results, the two assays were in agreement 99.8% (597/598) of the time. A summary of the combined qualitative and quantitative *Salmonella* results can be found in Table 1. Briefly, the majority of samples were reported to be either below the LOQ and negative (n = 511) or below the LOQ and positive (n = 80). Only seven samples yielded quantitative results above the LOQ, with six of those being reported as positive and one as negative. Finally, the remaining samples (n = 71) yielded invalid quantitative results, either as a result of assay inhibition (n = 68) or laboratory error (n = 3), with the majority of those being positive for *Salmonella*.

Salmonella contamination was observed to vary across the three harvest facilities and nine sampling locations. A portion of this overall variance was attributed to the repeated measure of individual carcass, specifically 13.1% of the variance observed in the qualitative analysis. Although *Salmonella* prevalence and concentrations were numerically different at each of the three harvest facilities, similar trends were

Table 1
Summary of observed *Salmonella* prevalence and concentration results for pork carcass swabs from all three harvest facilities ($N = 669$)[a]

Observed Result		Reported Result		
Concentration (log_{10} CFU/swab)	Prevalence	Concentration (log_{10} CFU/swab)	Prevalence	N[b]
<1	Negative	<LOQ	Negative	511
<1	Positive	<LOQ	Positive	80
>1	Positive	Observed Result	Positive	6
>1	Negative	Observed Result	Positive[c]	1
Invalid Result[d]	Positive	Removed from analysis	Positive	61
Invalid Result	Negative	Removed from analysis	Negative	10

[a] Seven swabs were removed from both quantitative and qualitative analyses due to laboratory error.
[b] N represents the number of samples with the same combination of observed results.
[c] Results did not agree; therefore, the qualitative result was conservatively adjusted to positive for analyses.
[d] Invalid quantitative results caused by analytical ($n = 68$) or laboratory ($n = 3$) error were excluded from quantitative analyses but did not impact the qualitative analyses.

observed. Mainly at each of the three facilities, *Salmonella* prevalence was highest on incoming pigs and significantly reduced by the scalding, dehairing, singeing, polishing, and shaving processes. Although *Salmonella* prevalence fluctuated during harvest, a final prevalence of 4 to 8% was observed on carcasses entering the chiller. *Salmonella* concentrations were observed to be low at each of the three facilities, with the large majority of the samples reported to be below the LOQ (<1 log_{10} CFU/swab).

Salmonella concentrations. A total of 675 swab samples were collected to determine the quantitative level of *Salmonella* throughout pork harvest at three commercial processing facilities. After the removal of nine samples due to laboratory error, a total of 666 quantitative results were analyzed (Table 2). Of the swabs analyzed, 598 yielded valid quantitative results while 68 were reported as invalid, due to assay inhibition. The majority (94.1%) of invalid results were obtained from exsanguination swabs (A; $n = 66$), reducing the sample size at exsanguination to $n = 2$, $n = 1$, and $n = 7$, for facilities A, B, and C, respectively. The remaining invalid sample results were collected at the gambrel table (B; $n = 3$) and after splitting (G; $n = 1$). Excluding these invalid results, the quantitative and qualitative assay results agreed for all but one sample (99.8% agreement). In this instance, the sample was reported to contain 1.15 log_{10} CFU/swab of *Salmonella* despite being reported as negative by the qualitative assay. To represent the most conservative estimate of contamination, the qualitative result was adjusted to positive prior to analyses.

The overall level of *Salmonella* was low in each of the harvest facilities and at each sampling locations. Of the 598 samples that yielded valid quantitative results, 591 were reported below the LOQ (≤1 log_{10} CFU/swab). The seven samples reported above the LOQ were collected from facility B ($n = 5$) and C ($n = 2$) and ranged in

Table 2
Distribution of quantitative *Salmonella* results from each of the three pork harvest facilities (A, B, and C) at each of the nine sampling locations ($N = 666$)

		Sample Distribution (n)[a]		
Harvest Facility	Sampling Location	< LOQ[b]	> LOQ[c] (range log_{10} CFU/swab)	Invalid[d]
A	Exsanguination	2	N/A	23
A	Gambrel Table	23	N/A	2
A	Shavers	25	N/A	N/A
A	Bunging	25	N/A	N/A
A	Opening	25	N/A	N/A
A	Evisceration	25	N/A	N/A
A	Split	25	N/A	N/A
A	Prewash Cabinet	25	N/A	N/A
A	Postwash Cabinet	25	N/A	N/A
B	Exsanguination	1	N/A	23
B	Gambrel Table	20	4 (1.43–1.56)	N/A
B	Shavers	25	N/A	N/A
B	Opening	24	N/A	N/A
B	Bunging	25	1 (1.20)	N/A
B	Evisceration	24	N/A	N/A
B	Split	23	N/A	1
B	Prewash Cabinet	24	N/A	N/A
B	Postwash Cabinet	24	N/A	N/A
C	Exsanguination	7	N/A	18
C	Gambrel Table	24	N/A	1
C	Shavers	24	1 (1.26)	N/A
C	Opening	25	N/A	N/A
C	Bunging	24	N/A	N/A
C	Evisceration	23	1 (1.15)	N/A
C	Split	25	N/A	N/A
C	Prewash Cabinet	25	N/A	N/A
C	Postwash Cabinet	24	N/A	N/A

[a] n represents the number of quantitative results in each category.
[b] Results below the limit of quantification (≤1 log_{10} CFU/swab).
[c] Results above the limit of quantification (>1 log_{10} CFU/swab).
[d] Invalid results represent though samples that were unable to be quantified due to assay inhibition.

concentration from 1.15 to 1.56 $\log_{10}$ CFU/swab. All of the swabs collected in facility A were reported to be below the LOQ. The highest concentration of *Salmonella* was observed at the gambrel table (B) in facility B, where four samples were reported above the LOQ ranging from 1.43 to 1.56 $\log_{10}$ CFU/swab. Other locations in which the concentration of *Salmonella* exceeded the LOQ included after bunging (D; 1.20 $\log_{10}$ CFU/swab) in facility B and after shavers (C; 1.26 $\log_{10}$ CFU/swab) and evisceration (F; 1.15 $\log_{10}$ CFU/swab) in facility C.

***Salmonella* prevalence.** The prevalence of *Salmonella* was impacted by harvest facility ($p = 0.006$), sampling location ($p < 0.001$), and the interaction between the two ($p = 0.004$; $n = 669$; Figure 2; Table 3). The prevalence of *Salmonella* on incoming pigs, sampled at exsanguination (A), ranged from 84% in facility A to 100% in facility C. The process of scalding and dehairing resulted in the first and one of the greatest reductions in *Salmonella* prevalence from exsanguination (A) to gambrel table (B), reducing the prevalence from 84 to 44%, 96 to 63%, and 100 to 32% in facilities A, B, and C, respectively ($p < 0.002$). The second reduction in *Salmonella* prevalence was observed between the gambrel table (B) and shavers (C); however, this reduction was only significant in facilities A and B, where the prevalence was reduced from 44 to 0% and 63 to 12%, respectively ($p < 0.001$).

After the shavers (C), more variability in *Salmonella* prevalence was observed between harvest facilities. The *Salmonella* prevalence in facility A remained extremely low for the remainder of processing, with the only positive sample being reported at postwash cabinet (I), increasing the prevalence from 0 to 4%. Facility B exhibited a slightly higher prevalence ranging from 0 to 21% from shavers (C) to the end of processing. Although the *Salmonella* population remained relatively constant during the later stages of processing, there were two instances in which the prevalence increased numerically: (1) after evisceration (E to F) and (2) after trimming immediately prior to the prewash cabinet (G to H). The increase in prevalence from 0 to 21% after trimming was unique to facility B; however, the increase observed after evisceration was observed in both facility B and facility C, increasing prevalence from 4 to 8% and 8 to 24%, respectively. The final step in the harvest process was an antimicrobial (PAA) wash cabinet. In facilities B and C, this intervention reduced the prevalence from 21 to 8% and 16 to 4%, respectively. However, in facility A, an increase in *Salmonella* prevalence was observed after the wash cabinet, after one carcass was identified as positive (0 to 4%). Despite the significant reductions in *Salmonella* prevalence achieved throughout the harvest process, *Salmonella* was identified on at least one carcass entering the chiller in each facility.

Discussion

Salmonella contamination continues to be an ongoing challenge for the meat and poultry industry (U.S. Department of Health and Human Services, 2022). With pork ranking among the top five commodity groups attributed to foodborne *Salmonella* illness and the anticipation of impending regulatory standards (87 FR 8774), many in the pork industry have focused their efforts on understanding and controlling *Salmonella* contamination within harvest and fabrication facilities (Interagency Food Safety Analytics Collaboration, 2024). Quantitative and qualitative *Salmonella* bio-maps, such as those generated in the current study, provide valuable insight related to contamination pathways and the efficacy of intervention strategies. This information can then be used by processors to make targeted, data-driven decisions to improve the safety of their products. Quantitative analysis determined that the levels of *Salmonella* were low throughout processing; however, the technology was limited in its ability to detect *Salmonella* in swabs with a high amount of organic material or background microbiota. Nonetheless, this study revealed that the greatest reductions in *Salmonella* prevalence were achieved during the initial stages of harvest, after either scalding and dehairing or singeing and polishing. Varia-

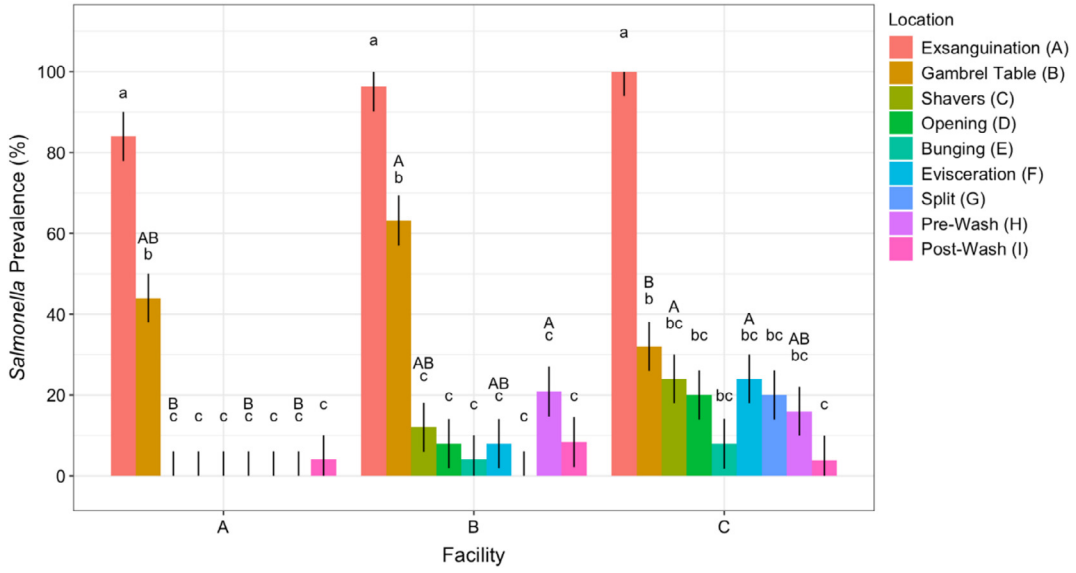


Figure 2. *Salmonella* prevalence by sample location at each harvest facility. Carcass swabs were collected from three commercial processing facilities (A, B, and C), five times a day for five days each at nine locations throughout harvest. At each sampling interval, a carcass was identified at exsanguination (A) and subsequently tracked through processing. Following exsanguination (A), carcasses were subject to scalding and dehairing before being rehung at the gambrel table (B). After rehanging, carcasses were singed, polished, manually shaved (C), opened (D), bunged (E), eviscerated (F), and split (G). Split carcasses were then manually trimmed and swabbed prior (H) to and after (I) the antimicrobial wash cabinet. Sponge-stick swabs were used to swab the ham, shoulder, and jowl of each carcass, and *Salmonella* prevalence (% positive) was assessed using the Gene-Up® system. The bars represented the estimated marginal means for *Salmonella* prevalence and with standard error bars ($N = 669$). Different lowercase letters (a, b, c) denote significant differences in *Salmonella* prevalence between sampling locations within each harvest facility, and different uppercase letters (A, B) denote significant differences in *Salmonella* prevalence between harvest facilities at individual sampling locations ($p \leq 0.05$). Harvest facility ($p = 0.006$), sampling location ($p < 0.001$), and the interaction between the two ($p = 0.004$) impacted *Salmonella* prevalence. *Salmonella* prevalence was highest on incoming pigs and largely reduced by scalding, dehairing, singeing, and polishing.

Table 3
Salmonella prevalence at each sampling location throughout processing at each pork harvest facility ($N = 669$)[1]. Harvest facility ($p = 0.006$), sampling location ($p < 0.001$), and the interaction between the two ($p = 0.004$) impacted *Salmonella* prevalence

Harvest Facility	Sampling Location	*Salmonella* Prevalence % (Positive/*n*)
A	Exsanguination	84% (21/25) [a]
A	Gambrel Table	44% (11/25) [b, AB]
A	Shavers	0% (0/25) [c, B]
A	Bunging	0% (0/25) [c]
A	Opening	0% (0/25) [c]
A	Evisceration	0% (0/25) [c, B]
A	Split	0% (0/25) [c]
A	Prewash Cabinet	0% (0/25) [c, B]
A	Postwash Cabinet	4% (1/25) [c]
B	Exsanguination	96% (23/24) [a]
B	Gambrel Table	63% (15/24) [b, A]
B	Shavers	12% (3/25) [c, AB]
B	Opening	8% (2/25) [c]
B	Bunging	4% (1/25) [c]
B	Evisceration	8% (2/25) [c, AB]
B	Split	0% (0/25) [c]
B	Prewash Cabinet	21% (5/24) [c, A]
B	Postwash Cabinet	8% (2/24) [c]
C	Exsanguination	100% (25/25) [a]
C	Gambrel Table	32% (8/25) [b, B]
C	Shavers	24% (6/25) [bc, A]
C	Opening	20% (5/25) [bc]
C	Bunging	8% (2/24) [bc]
C	Evisceration	24% (6/25) [bc, A]
C	Split	20% (5/25) [bc]
C	Prewash Cabinet	16% (4/25) [bc, AB]
C	Postwash Cabinet	4% (1/24) [c]

Values with different lowercase letters (a-b) denote statistically significant differences between sampling locations within each harvest facility. Values with different uppercase letters (A-B) denote statistically significant differences between harvest facilities within each sampling location.

[1] N represents the total number of qualitative results considered in the analysis.

tions between the three harvest facilities highlighted the value of individualized bio-mapping and encouraged the sharing of information between establishments. Finally, the detection of *Salmonella* on carcasses entering the chiller after harvest suggested that either some populations of *Salmonella* persisted through processing or were reintroduced through cross-contamination in the final stages of harvest, ultimately resulting in a potential food safety risk.

Salmonella quantitation has been one of the latest technological advancements to make its mark on the food industry. Customer demand has led several diagnostic companies to develop their own testing platforms; however, a recent study revealed that not all assays perform equally (Schmidt et al., 2024). These results support the need for further investigation regarding the accuracy and sensitivity of molecular-based detection systems when applied to the food industry. The efficacy and sensitivity of these assays become exceedingly important when considering their potential use by regulatory agencies to assess compliance. In the current study, an enrichment-free qPCR-based platform was used due to its rapid time to results (<4 h), ease of sample processing, and accuracy; however, like other platforms that employ nonenriched molecular quantification, the sensitivity (>1 $\log_{10}$ CFU) and susceptibility to environmental interference create a challenge when applying this technology to real-world processing environments (Bell et al., 2016; Schmidt et al., 2024; Yang et al., 2025). The ability to generate quantitative data in real-time provides processer with an unparalleled level of control over their process; however, assay sensitivity and robustness need to be considered as they are key to the generation of actionable data.

Few studies have aimed to quantify the level of natural *Salmonella* contamination during pork harvest; those that have report the levels to be at or below 2 $\log_{10}$ CFU/g, not unlike other commodity groups (Bell et al., 2016; Berends et al., 1996; Bueno López et al., 2022). *Salmonella* contamination in the current study was detected at slightly lower concentrations, with the greatest value being reported at 1.56 $\log_{10}$ CFU/swab. Unfortunately, with the large majority of the swabs reported below the LOQ (<1 $\log_{10}$ CFU/swab), few comparisons could be made. This study aimed to quantify the incoming load of *Salmonella* by collecting carcass swabs immediately prior to exsanguination; however, this aim went largely unmet due to the high number of invalid results obtained due to assay interference. This interference was likely caused by excess soil/feces, known PCR inhibitors, which were visibly present in high quantities on carcasses at exsanguination (Sidstedt et al., 2020). Although similar studies have chosen to avoid this analytical challenge by collecting the first swab sample at the gambrel table, after scalding and dehairing had removed the majority of environmental inhibitors, this too has its limitations (Bueno López et al., 2022). Understanding the level of incoming *Salmonella* contamination is necessary to evaluate the efficacy of pre- and postharvest food safety intervention strategies; therefore, alternative sample processing strategies should be further explored. The quantitative data collected in the current study supported the existing theory that *Salmonella* contamination levels are low during pork harvest, contrary to what has been predicted through some quantitative microbial risk assessments (Ntakiyisumba et al., 2023).

To improve the resolution of quantitative analysis, the paired qualitative results were leveraged to identify the presence/absence of *Salmonella* at a LOD of 1 CFU/swab, similar to previous studies (Bueno López et al., 2022). The quantitative assay lacked the ability to identify negative samples (<1 CFU/swab) and reported all samples below the LOQ as <1 $\log_{10}$ CFU/swab. Utilizing the paired results, samples were categorized into three main groups: (1) below the LOQ and negative, (2) below the LOQ and positive, and (3) above the LOQ and positive. Samples below the LOQ and negative would be considered true negatives. Those samples below the LOQ and positive would be considered low concentration positives, and those above the LOQ and positive would be considered high concentration positives. Although the quantitative results were skewed by the large number of invalid results at exsanguination, the concentration data follow a similar pattern to what was observed with *Salmonella* prevalence. The highest concentration of *Salmonella* was observed at the gambrel table, followed closely by samples collected after the shavers, after bunging, and after evisceration. Because prevalence is often assumed to be correlated with concentration, the concentration at exsanguination was expected to be several logarithmic units higher than what was reported in this study. However, the majority of the evidence suggests that *Salmonella* concentrations do not exceed 2 to 3 $\log_{10}$ CFU/swab on live pigs and pig carcasses at exsanguination (Bueno López et al., 2022; van Hoek et al., 2012). While it is more desirable to have lower levels of *Salmonella* contamination during pork harvest, low concentrations of *Salmonella* still pose a risk in the case of highly virulent strains and/or improperly handled foods (Fluit, 2005; Lamichhane et al., 2024; Redmond & Griffith, 2003). Optimization of sample processing and molecular sensitivity will be key to improving the commercial utility of quantitative detection technologies in the food industry.

Salmonella prevalence in the pork supply chain has been extensively researched (Spescha et al., 2006; U.S. Department of Agriculture Food Safety and Inspection Service, 2011; Wheatley et al., 2014; Wottlin et al., 2022, 2024). Most of the existing literature agrees that *Salmonella* prevalence is highest at the beginning of harvest and reduced most significantly during the initial stages of processing (Bueno López et al., 2022; U.S. Department of Agriculture Food Safety and Inspection Service, 2011). *Salmonella* prevalence in the current study ranged from 86 to 100% at exsanguination, approximately 20% higher than the prevalence reported at the gambrel table by Bueno López et al. (2022) and at preevisceration by USDA FSIS (2011).

After exsanguination, pork carcasses were subject to scalding and dehairing, a process that has been categorized as both an intervention

step, due to the application of hot water, and a potential source of cross-contamination (Hdaifeh et al., 2020). In the current study, as well as several others, *Salmonella* prevalence was significantly reduced after scalding and dehairing (Gill et al., 1995; Wheatley et al., 2014). Two methods of scalding were evaluated in the current study, steam scalding in facility A and immersion scalding in facility B and C. Steam scalding or vertical scalding is less common in U.S. pork harvest facilities as compared to immersion scalding; however, in recent years, it has grown in popularity due to its improved hygiene and energy efficiency (U.S. Department of Agriculture Food Safety and Inspection Service, 2023). Although the scalding method was not directly correlated with an increased reduction in *Salmonella* prevalence, facility A, which utilized a steam scalding system, did exhibit the lowest overall *Salmonella* prevalence from gambrel table to postwash cabinet, suggesting that the scalding method may impact *Salmonella* prevalence in later stages of processing.

After carcasses were rehung at the gambrel table, they were subsequently singed and polished to remove any remaining hair, resulting in another reduction in *Salmonella* prevalence. While similar results have been reported in other bio-mapping studies, others reported that polishing can increase external contamination and recommend it be replaced with a hot water spray or pasteurization step (Bueno López et al., 2022; Spescha et al., 2006). The current study demonstrated that external decontamination during the initial stages of processing, if performed properly, does significantly reduce the prevalence of *Salmonella* throughout the remainder of harvest.

Effective sanitary dress procedures play a crucial role in reducing the level of *Salmonella* contamination during harvest (U.S. Department of Agriculture Food Safety and Inspection Service, 2023). The process of bunging, eviscerating, and splitting has been associated with increases in *Salmonella* prevalence and therefore crucial steps in which process controls can be applied to minimize risk (Bueno López et al., 2022; O'Connor et al., 2012; Spescha et al., 2006). In the current study, we observed slight increases in *Salmonella* prevalence after evisceration in facilities B and C, indicating some level of cross-contamination or direct contact with contaminated ingesta or fecal material during the removal process. While the majority of pork processing is still dependent upon a skilled labor force, recent improvements in automation have enabled the development of robotic arms capable of opening and splitting carcasses (Kim et al., 2023). Interestingly, while no direct correlations were made between automation and *Salmonella* prevalence, in the current study, the facilities (B and C) that employed robot technology to open and split carcasses also had numerically higher prevalence rates. In the future, as robot technology continues to expand to overcome labor shortages, it will be crucial to consider their impacts on food safety.

In the final stages of processing, carcasses were trimmed and treated with an antimicrobial spray. Carcass trimming was performed manually, creating numerous cross−contamination opportunities, as evidenced by the increased prevalence in facilities B. The fact that these increases were not observed in facilities A and C could be related to the overall prevalence; however, it could also indicate the need for additional intervention strategies to reduce the potential for cross-contamination. A quantitative microbial risk assessment conducted by Barron et al. (2009) estimated that the probability of cross-contamination during splitting and trimming could be up to 50%. As a final hurdle to reduce contamination, carcasses were subject to a PAA wash prior to chilling. Organic acid washes are well documented to reduce bacterial contamination on carcasses and, for that reason, are frequently incorporated as interventions in harvest and fabrication facilities (Food and Agriculture Organization of the United Nations & World Health Organization, 2016; Gonzalez et al., 2023). In the current study, a reduction in *Salmonella* prevalence was observed after the final wash, as expected, in facilities B and C; however, in facility A, an increase in prevalence was observed as a result of one positive sample being detected postwash cabinet. Although the increased prevalence in facility A after the wash cabinet was unexpected, similar results have been reported in other studies (Bueno López et al., 2022). The most probable explanation for these results is that *Salmonella* was introduced during an earlier stage of processing and, in turn, managed to persist and avoid detection up until the final sampling location.

After the harvest process is complete, pork carcasses must be chilled to an internal temperature below 4.4 °C. Chilling has been shown to have a beneficial effect on carcass contamination, reducing the probability of *Salmonella* prevalence; however, depending on the method of chilling and packing density of the chiller, if carcasses remain warm and moist, bacterial growth has been shown to occur (Reid et al., 2017). The results from the current study highlight the importance of understanding *Salmonella* survival and potential growth kinetics under various chilling conditions, as it is evident that *Salmonella* was present on at least a portion of the carcasses entering the chiller.

Conclusions

The results demonstrate the value that can be gained through the generation of quantitative and qualitative *Salmonella* bio-maps of commercial pork harvest facilities. The quantitative results provided evidence that the level of *Salmonella* contamination throughout harvest is low; however, the technology was limited in its level of sensitivity and susceptibility to environmental factors, ultimately reducing its commercial utility. *Salmonella* prevalence data revealed that the majority of pigs entering the harvest facility were positive for *Salmonella*; however, this prevalence was significantly reduced immediately following exsanguination. The poor correlation between *Salmonella* contamination and individual processes or equipment highlights the presence of systematic rather than localized contamination. Finally, the identification of *Salmonella* at every stage throughout the harvest process demonstrates the urgent need for implementation of improved detection and process control strategies.

CRediT authorship contribution statement

J.A. Brown: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **S.C. Ricke:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jessica Brown reports that financial support was provided by industry partners and the University of Wisconsin Foundation through the Bray-Woodbury Wisconsin Distinguished Graduate Fellowship Fund in Meat Science. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank our industry collaborator for providing commercial samples, laboratory access, and financial support throughout this study. Author J.A.B. would like to acknowledge the University of Wisconsin Foundation at the University of Wisconsin-Madison for its continued support through the Bray-Woodbury Wisconsin Distinguished Graduate Fellowship Fund in Meat Science. The authors declare that the research was conducted in the absence of any commercial or financial relationship that could be construed as a potential conflict of interest.

References

Barron, U. G., Soumpasis, I., Butler, F., Prendergast, D., Duggan, S., & Duffy, G. (2009). Estimation of prevalence of *Salmonella* on pig carcasses and pork joints, using a quantitative risk assessment model aided by meta-analysis. *Journal of Food Protection, 72*(2), 274–285. https://doi.org/10.4315/0362-028X-72.2.274.

Bell, R. L., Jarvis, K. G., Ottesen, A. R., McFarland, M. A., & Brown, E. W. (2016). Recent and emerging innovations in *Salmonella* detection: A food and environmental perspective. *Microbial Biotechnology, 9*(3), 279–292. https://doi.org/10.1111/1751-7915.12359.

Berends, B. R., Urlings, H. A. P., Snijders, J. M. A., & Van Knapen, F. (1996). Identification and quantification of risk factors animal management and transport regarding in *Salmonella* spp. in pigs. *International Journal of Food Microbiology, 30*(1–2), 37–53. https://doi.org/10.1016/0168-1605(96)00990-7Getrightsandcontent.

Bueno López, R., Vargas, D. A., Jimenez, R. L., Casas, D. E., Miller, M. F., Brashears, M. M., & Sanchez-Plata, M. X. (2022). Quantitative bio-mapping of *Salmonella* and indicator organisms at different stages in a commercial pork processing facility. *Foods, 11*(17), 2580. https://doi.org/10.3390/foods11172580.

Centers for Disease Control and Prevention (2019). *Antibiotic resistance threats in the United States, 2019*. U.S. Department of Health and Human Services, CDC. https://doi.org/10.15620/cdc:82532.

Fluit, A. C. (2005). Towards more virulent and antibiotic-resistant *Salmonella*? *FEMS Immunology & Medical Microbiology, 43*(1), 1–11. https://doi.org/10.1016/j.femsim.2004.10.007.

Food and Agriculture Organization of the United Nations, & World Health Organization. (2016). *Interventions for the control of non-typhoidal Salmonella spp. in beef and pork*. www.fao.org/food/food-safety-quality.

Gill, C. O., McGinnis, D. S., Bryant, J., & Chabot, B. (1995). Decontamination of commercial, polished pig carcasses with hot water. *Food Microbiology, 12*, 143–149. https://doi.org/10.1016/S0740-0020(95)80090-5.

Gonzalez, S. V., Nair, M. N., Belk, K. E., & Geornaras, I. (2023). Efficacy of antimicrobial spray treatments in reducing *Salmonella enterica* populations on chilled pork. *Journal of Food Protection, 86*(4)100068. https://doi.org/10.1016/j.jfp.2023.100068.

Gragg, S. E., & Brashears, M. M. (2014). Microbiological safety of meat | Hurdle technology. In M. Dikeman & C. Devine (Eds.), *Encyclopedia of Meat Sciences* (2nd ed., pp. 345–347). Academic Press. https://doi.org/10.1016/B978-0-12-384731-7.00134-3.

Hdaifeh, A., Khalid, T., Boué, G., Cummins, E., Guillou, S., Federighi, M., & Tesson, V. (2020). Critical analysis of pork QMRA focusing on slaughterhouses: Lessons from the past and future trends. *Foods, 9*(11), 1704. https://doi.org/10.3390/foods9111704.

Interagency Food Safety Analytics Collaboration (2024). *Foodborne illness source attribution estimates for Salmonella, Escherichia coli* O157, and *Listeria monocytogenes – United States, 2022*.

Kim, J., Kwon, Y. K., Kim, H. W., Seol, K. H., & Cho, B. K. (2023). Robot technology for pork and beef meat slaughtering process: A review. *Animals, 13*(4), 651. https://doi.org/10.3390/ani13040651.

Kothary, M. H., & Babu, U. S. (2001). Infective dose of foodborne pathogens in volunteers: A review. *Journal of Food Safety, 21*(1), 49–68. https://doi.org/10.1111/j.1745-4565.2001.tb00307.x.

Lamichhane, B., Mawad, A. M. M., Saleh, M., Kelley, W. G., Harrington, P. J., Lovestad, C. W., Amezcua, J., Sarhan, M. M., el Zowalaty, M. E., Ramadan, H., Morgan, M., & Helmy, Y. A. (2024). Salmonellosis: An overview of epidemiology, pathogenesis, and innovative approaches to mitigate the antimicrobial resistant infections. *Antibiotics, 13*(1), 76. https://doi.org/10.3390/antibiotics13010076.

Li, H., Wang, H., D'Aoust, J.-Y., & Maurer, J. (2012). Salmonella species. In M. P. Doyle & R. L. Buchanan (Eds.), *Food Microbiology: Fundamentals and Frontiers* (4th ed., pp. 223–261). . https://doi.org/10.1128/9781555818463.

Neyaz, L. A., Alghamdi, H. S., Alghashmari, R. M., Alswat, S. S., Almaghrabi, R. O., Bazaid, F. S., Albarakaty, F. M., Elbanna, K., & Abulreesh, H. H. (2024). A comprehensive review on the current status of culture media for routine standardized isolation of *Salmonella* and *Shigella* spp. from contaminated food. *Journal of Umm Al-Qura University for Applied Sciences*. https://doi.org/10.1007/s43994-024-00205-2.

Ntakiyisumba, E., Lee, S., & Won, G. (2023). Identification of risk profiles for *Salmonella* prevalence in pig supply chains in South Korea using meta-analysis and a quantitative microbial risk assessment model. *Food Research International, 170*, 112999. https://doi.org/10.1016/j.foodres.2023.112999.

O'Connor, A. M., Wang, B., Denagamage, T., & McKean, J. (2012). Process mapping the prevalence of *Salmonella* contamination on pork carcass from slaughter to chilling: A systematic review approach. *Foodborne Pathogens and Disease, 9*(5), 386–395. https://doi.org/10.1089/fpd.2011.1040.

Park, S. H., Aydin, M., Khatiwara, A., Dolan, M. C., Gilmore, D. F., Bouldin, J. L., Ahn, S., & Ricke, S. C. (2014). Current and emerging technologies for rapid detection and characterization of *Salmonella* in poultry and poultry products. *Food Microbiology, 38*, 250–262. https://doi.org/10.1016/j.fm.2013.10.002.

Priyanka, B., Patil, R., & Dwarakanath, S. (2016). A review on detection methods used for foodborne pathogens. *Indian Journal of Medical Research, 144*(3), 327–338. https://doi.org/10.4103/0971-5916.198677.

Redmond, E. C., & Griffith, C. J. (2003). Consumer food handling in the home: A review of food safety studies. *Journal of Food Protection, 66*(1), 130–161. https://doi.org/10.4315/0362-028X-66.1.130.

Reid, R., Fanning, S., Whyte, P., Kerry, J., Lindqvist, R., Yu, Z., & Bolton, D. (2017). The microbiology of beef carcasses and primals during chilling and commercial storage. *Food Microbiology, 61*, 50–57. https://doi.org/10.1016/j.fm.2016.08.003.

Scallan, E., Hoekstra, R. M., Angulo, F. J., Tauxe, R. V., Widdowson, M.-A., Roy, S. L., Jones, J. L., & Griffin, P. M. (2011). Foodborne illness acquired in the United States—major pathogens. *Emerging Infectious Diseases, 17*(1), 7–15. https://doi.org/10.3201/eid1701.P11101.

Schmidt, J. W., Carlson, A., Bosilevac, J. M., Harhay, D., Arthur, T. M., Brown, T., Wheeler, T. L., & Vipham, J. L. (2024). Evaluation of methods for identifying poultry wing rinses with *Salmonella* concentrations greater than or equal to 10 CFU/mL. *Journal of Food Protection, 87*(11)100362. https://doi.org/10.1016/j.jfp.2024.100362.

Searle, S. R., Speed, F. M., & Milliken, G. A. (1980). Population marginal means in the linear model: An alternative to least squares means. *The American Statistician, 34*(4), 216–221. https://doi.org/10.1080/00031305.1980.10483031.

Sidstedt, M., Rådström, P., & Hedman, J. (2020). PCR inhibition in qPCR, dPCR and MPS—mechanisms and solutions. *Analytical and Bioanalytical Chemistry, 412*(9), 2009–2023. https://doi.org/10.1007/s00216-020-02490-2.

Spescha, C., Stephan, R., & Zweifel, C. (2006). Microbiological contamination of pig carcasses at different stages of slaughter in two European Union–approved abattoirs. *Journal of Food Protection, 69*(11), 2568–2575. https://doi.org/10.4315/0362-028X-69.11.2568.

R Core Team (2024). R: a language and environment for statistical computing. *R foundation for statistical computing*. https://www.R-project.org/.

U.S. Department of Agriculture Food Safety and Inspection Service (2005). *Guidelines for Escherichia coli testing for process control verification in cattle and swine slaughter establishments*. https://www.fsis.usda.gov/guidelines/1996-0001.

U.S. Department of Agriculture Food Safety and Inspection Service (2011). *FSIS nationwide market hog microbiological baseline*. https://www.fsis.usda.gov/sites/default/files/media_file/2020-07/Baseline_Data_Market_Hogs_Study_Design.pdf.

U.S. Department of Agriculture Food Safety and Inspection Service (2013). *Salmonella action plan*. https://www.fsis.usda.gov/sites/default/files/media_file/2020-10/SAP-120413.pdf.

U.S. Department of Agriculture Food Safety and Inspection Service (2019). *Verifying swine slaughter establishments maintain adequate procedures for preventing contamination of carcasses and parts by enteric pathogens*. Directive. 6410.4. https://www.fsis.usda.gov/sites/default/files/media_file/2020-07/6410.4.pdf.

U.S. Department of Agriculture Food Safety and Inspection Service (2021). *Foodborne pathogen test kits validated by independent organizations*. https://www.fsis.usda.gov/sites/default/files/media_file/2021-10/validated-test-kit.pdf.

U.S. Department of Agriculture Food Safety and Inspection Service (2023). *FSIS guideline to control Salmonella in swine slaughter and pork processing establishments*. https://www.fsis.usda.gov/guidelines/2023-0003.

U.S. Department of Agriculture Food Safety Inspection Service (2018). Changes to *Salmonella* and *Campylobacter* verification testing program: Revised categorization and follow-up sampling procedures. *Federal Register, 83*, 56046–56049.

U.S. Department of Agriculture Food Safety Inspection Service (2019a). Changes to the *Salmonella* verification testing program: Proposed performance standards for *Salmonella* in raw ground beef and beef manufacturing trimmings and related agency verification procedures. *Federal Register, 84*, 57688–57695.

U.S. Department of Agriculture Food Safety Inspection Service (2022). Changes to the *Salmonella* verification testing program: Proposed performance standards for *Salmonella* in raw comminuted pork and intact or non-intact pork cuts and related agency verification procedures. *Federal Register, 87*, 8774–8780.

U.S. Department of Agriculture Food Safety Inspection Service (1996). Pathogen reduction; hazard analysis and critical control point (HACCP) systems; final rule. 9 CFR 304, 308, 310, 320, 327, 381, 416, and 417. *Federal Register, 61*, 38806–38989.

U.S. Department of Agriculture Food Safety Inspection Service (2016). New performance standards for *Salmonella* and *Campylobacter* in not-ready-to-eat comminuted chicken and turkey products and raw chicken parts and changes to related agency verification procedures: response to comments and announcement of implementation schedule. *Federal Register, 81*, 7285–7300.

U.S. Department of Health and Human Services (2022). *Reduce infections caused by Salmonella* - FS-04. Retrieved October 15, 2025, from: https://odphp.health.gov/healthypeople/objectives-and-data/browse-objectives/foodborne-illness/reduce-infections-caused-salmonella-fs-04.

van Hoek, A. H. A. M., de Jonge, R., van Overbeek, W. M., Bouw, E., Pielaat, A., Smid, J. H., Malorny, B., Junker, E., Löfström, C., Pedersen, K., Aarts, H. J. M., & Heres, L. (2012). A quantitative approach towards a better understanding of the dynamics of *Salmonella* spp. in a pork slaughter-line. *International Journal of Food Microbiology, 153*(1–2), 45–52. https://doi.org/10.1016/j.ijfoodmicro.2011.10.013.

Wheatley, P., Giotis, E. S., & McKevitt, A. I. (2014). Effects of slaughtering operations on carcass contamination in an Irish pork production plant. *Irish Veterinary Journal, 67*(1), 1. https://doi.org/10.1186/2046-0481-67-1.

World Health Organization (2018). *Salmonella* (non-typhoidal). *Fact Sheets*.

Wottlin, L. R., Harvey, R. B., Norman, K. N., Burciaga, S., Loneragan, G. H., Droleskey, R. E., & Anderson, R. C. (2022). Prevalence and antimicrobial resistance of nontyphoidal *Salmonella enterica* from head meat and trim for ground product at pork processing facilities. *Journal of Food Protection, 85*(7), 1008–1016. https://doi.org/10.4315/JFP-22-049.

Wottlin, L. R., Harvey, R. B., Norman, K. N., Droleskey, R. E., Andrews, K., Jackson, S. J., Anderson, R. C., & Poole, T. L. (2024). Prevalence and characterization of *Salmonella* during pork sausage manufacturing. *Microorganisms, 12*(8), 1599. https://doi.org/10.3390/microorganisms12081599.

Yang, Q., Zu, J., Zhang, S., Liu, C., Qin, X., & Xu, W. (2025). An overview of rapid detection methods for *Salmonella*. *Food Control, 167*, 110771. https://doi.org/10.1016/j.foodcont.2024.110771.